

cell0_recip_stack Native Galaxy Derivation

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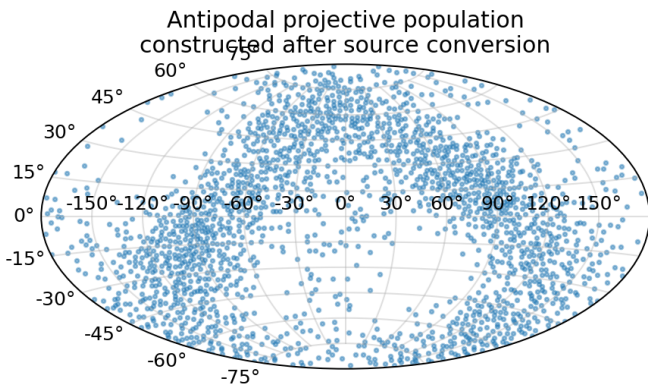
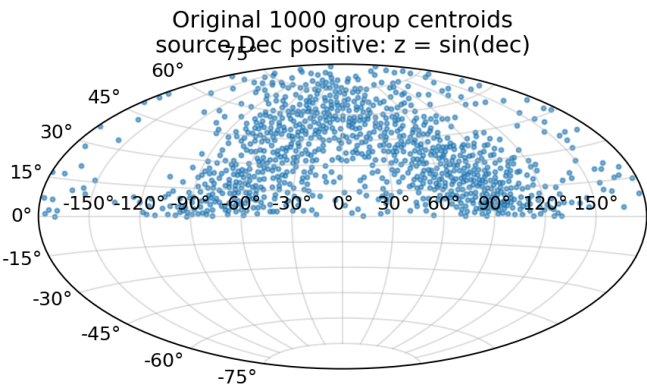
Strict provenance rule

This paper derives a native galaxy-sourced cell0 object. Authoritative cell3 is not used in this derivation. The allowed path is: galaxy-axis unit sphere -> method-specific discovery evidence -> native cell0/original rhombohedral scaffold -> computed geometry. The perturbation to cell3 is handled only in the separate bridge paper.

Source data and Dec convention audit

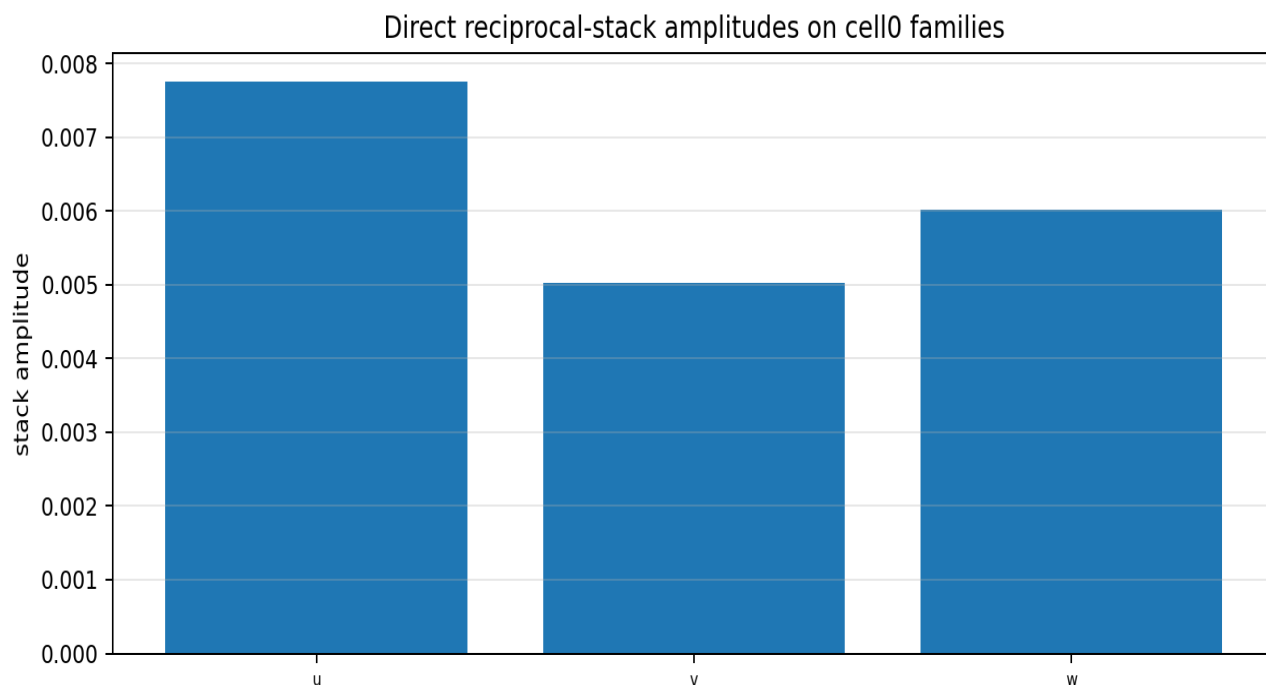
The source galaxy-axis group centroids are converted into unit vectors before any antipodal extension. This audit is a hard gate: all original source centroids have positive Dec and therefore positive z. If this plot or table is inverted, the entire derivation is invalid.

Audit item	Value
source file	galaxy_map(29).txt
source data rows	126006
unique group centroids	1000
Dec range	0.088361 to 87.522719 deg
z range before antipodes	0.001542190338 to 0.999065438844
RA/Dec round-trip error	RA 8.527e-14 deg; Dec 4.263e-14 deg
conversion	$x=\cos(\text{dec})\cos(\text{ra})$, $y=\cos(\text{dec})\sin(\text{ra})$, $z=\sin(\text{dec})$
antipodal construction	$X = \{x_i, -x_i\}$, performed after source conversion



Method-specific discovery evidence

The direct reciprocal-stack route begins with candidate plane-family normals and evaluates the scalar coordinates $s_i = n \cdot x_i$ for folded concentration under spacing Delta. It is the shortest derivation once the galaxy candidate normals have been identified. The following stack metrics are calculated directly from the corrected galaxy unit-sphere population.



Native cell0 reciprocal scaffold and edge matrix

The native cell0 object is the original galaxy-derived Lattice2-like rhombohedral cell. It is recovered before any cell2 or cell3 refinement is applied. The reciprocal scaffold is computed from the edge matrix by dual inversion: each row of A^{-1} gives a plane-family covector; normalizing that row gives the unit normal and its reciprocal norm gives the stack spacing Delta.

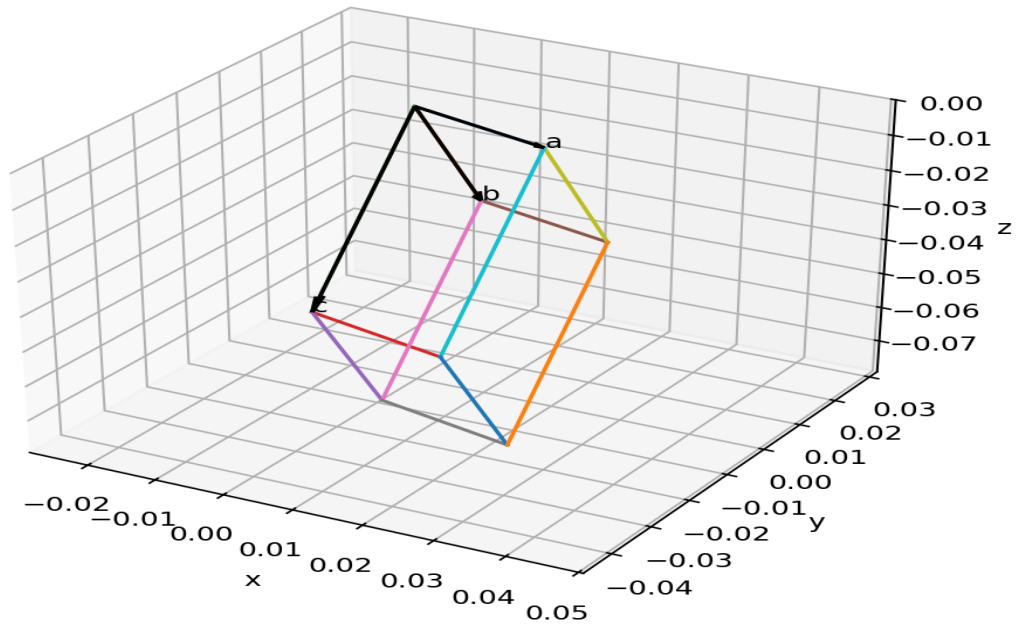
plane family vector, normalizing that row gives the unit normal and its reciprocal norm gives the stack spacing Δ (Å).

family	RA deg	Dec deg	Delta	stack amp	RMS/Delta
u	355.191046886673	1.870608720286	0.020500480431	0.007748189624	0.287072453261
v	84.135717852903	-29.42899034867	0.028100315543	0.00502400157	0.288739456951
w	258.000576203387	-21.00069260068	0.037700360495	0.00602075178	0.288459504875

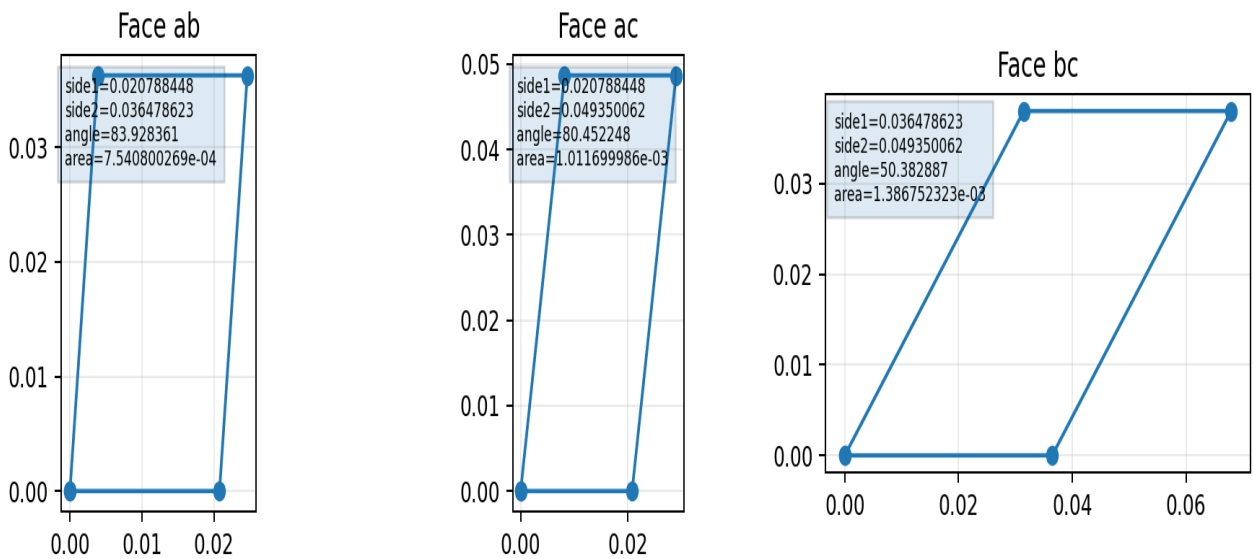
row	a column	b column	c column
1	0.020373	0.002201	-6.360000000000e-04
2	-0.003415	0.012896	-0.024293
3	-0.002332	-0.034052	-0.042952

quantity	cell0/original value
a	0.020788448186
b	0.036478622795
c	0.0493500623
theta_ab	83.928361201001
theta_ac	80.452247903308
theta_bc	50.382886569535
A_ab	7.540800268680e-04
A_ac	0.001011699986
A_bc	0.001386752323
volume	2.842908885472e-05
surface area	0.006305064672

Native galaxy-derived cell0 / original Lattice2 edge matrix



Native galaxy-derived cell0 faces



Native face-channel values computed from cell0

bridge	cell0 value
A_bc/A_ac	1.37071497604
A_bc/A_ab	1.838998877184
A_ac/A_ab	1.341634774063
eta_N	1.838998877184e+03
c_N	137.071497603998
alpha_N	0.007295462715

Recursive hostile audit

Gate	Status
source galaxy Dec not inverted	PASS
antipodal extension applied after correct unit vectors	PASS
native cell0 derived before cell3 comparison	PASS
authoritative cell3 not used in this derivation paper	PASS
ReportLab core fonts only	PASS
all table text wrapped in Paragraph cells	PASS